

MediKiosk — Deployment & Execution Guide

Step-by-Step Instructions for Online (Cloud) & Offline (Edge) Modes across macOS & Windows

System Overview: MediKiosk (SIH26047) contains a **FastAPI Backend (Port 8000)** and a **Vite Frontend Dev Server (Port 5173)**.

- **Online Mode:** Uses Google Gemini Flash (Vision/Reasoning) + Groq Llama-3.1-70B (Fast Dialogue) + Sarvam AI (Indic Speech).
- **Offline Mode:** Uses Local Ollama (Qwen-2.5 7B/3B) + Local RapidOCR + Offline Multilingual Embeddings + SQLite DB.

1. Prerequisites & System Setup

Ensure the following runtimes are installed on your machine:

Runtime / Dependency	Required Version	Installation Command / Notes
Python	3.10 to 3.12	Windows: python.org macOS: brew install python@3.11
Node.js & npm	Node 18+ (LTS)	Windows: nodejs.org macOS: brew install node
Ollama <i>(For Offline)</i>	Latest	Download from ollama.com (or brew install ollama)
Git	Latest	Windows: git-scm.com macOS: xcode-select --install

2. Initial Project Setup (One-Time)

Clone the repository, create a virtual environment, and install backend and frontend dependencies.

macOS / Linux Terminal:

```
# 1. Clone repo and navigate into directory
git clone https://github.com/Aaryan7117/pts.git
cd pts

# 2. Create and activate Python virtual environment
python3 -m venv venv
source venv/bin/activate

# 3. Install Python dependencies
pip install -r requirements.txt

# 4. Install Frontend dependencies
cd frontend && npm install && cd ..
```

Windows (PowerShell / Command Prompt):

```
# 1. Clone repo and navigate into directory
git clone https://github.com/Aaryan7117/pts.git
cd pts

# 2. Create and activate Python virtual environment
python -m venv venv
.\venv\Scripts\activate

# 3. Install Python dependencies
pip install -r requirements.txt

# 4. Install Frontend dependencies
cd frontend; npm install; cd ..
```

3. Running in ONLINE Mode (Cloud-Accelerated)

Online mode leverages **Gemini 2.5 Flash** for prescription handwriting vision extraction, **Groq Llama-3.1-70B** for conversational speed (<500ms), and **Sarvam AI** for high-fidelity Indian speech synthesis/recognition.

Step 1: Configure .env for Cloud Mode

```
DEPLOYMENT_MODE=CLOUD
GEMINI_API_KEY=AQ.Ab8RN6Lb5qAj7mSiImPHPGEQLQtJGNPq3AZvfK19yxBfxJ5BpA
GROQ_API_KEY=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
GROQ_MODEL=llama-3.1-70b-versatile
SARVAM_API_KEY=sk_2vbgua6g_fDCxR162SInIDZNYlcL037pF
```

Step 2: Start Services (macOS vs. Windows)

Option A: macOS / Linux (Two Terminal Tabs):

```
# Terminal Tab 1: Start FastAPI Backend
source venv/bin/activate
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload

# Terminal Tab 2: Start Frontend Dev Server
cd frontend
npm run dev -- --host 0.0.0.0 --port 5173
```

Option B: Windows (Automated Launcher or Manual PowerShell):

```
# Method 1: Automatic 1-Click Batch Runner (Launches both backend and frontend)
.\run_demo.bat

# Method 2: Manual PowerShell (Terminal 1 - Backend)
.\venv\Scripts\activate
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload

# Method 2: Manual PowerShell (Terminal 2 - Frontend)
cd frontend; npm run dev -- --host 0.0.0.0 --port 5173
```

4. Running in OFFLINE Mode (Zero-Cloud Edge Deployment)

Offline mode functions with **zero internet connection**, utilizing local Ollama models (Qwen-2.5 7B/3B), local RapidOCR for prescription text extraction, and CPU multilingual sentence embeddings.

Step 1: Install & Pull Ollama Models (One-Time Setup)

```
# 1. Start the Ollama background service
ollama serve

# 2. In another terminal, pull the required clinical models
ollama pull qwen2.5:7b
ollama pull qwen2.5:3b      # Optional fallback for lightweight CPUs
```

Step 2: Update .env for Standalone / Offline Mode

```
DEPLOYMENT_MODE=STANDALONE
OLLAMA_BASE_URL=http://localhost:11434/v1
OLLAMA_MODEL=qwen2.5:7b
OLLAMA_FALLBACK_MODEL=qwen2.5:3b
EMBEDDING_MODEL_NAME=sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2
EMBEDDING_WARMUP_ON_STARTUP=true
```

Step 3: Launch Local Services (macOS & Windows)

```
# 1. Verify Ollama is running at http://localhost:11434
curl http://localhost:11434/api/tags

# 2. Launch Backend (macOS: source venv/bin/activate | Win: .\env\Scripts\activate)
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload

# 3. Launch Frontend in a separate window
cd frontend && npm run dev -- --host 0.0.0.0 --port 5173
```

5. Demo Data Seeding & System Verification

To populate the system with synthetic patient cases, high-risk drug-drug interaction test cases, and sample prescriptions, execute the seed script:

```
# Populate database with realistic multilingual OPD cases & AYUSH interaction flags
python scripts/seed_demo_data.py
```

6. Portal Navigation & URL Directory

Once both services are running, open your web browser to access the following interfaces:

Interface / Portal	Access URL	Description & Credentials
Dual-Sided Web Hub	http://localhost:5173/?view=hub	Main control dashboard showing live kiosk and doctor views side-by-side.
Patient Voice Kiosk	http://localhost:5173/?view=patient or http://localhost:5173/	Multilingual touch/voice intake kiosk for patient registration and symptoms.
Doctor Workstation	http://localhost:5173/?view=doctor	Clinical dashboard with triage queue, gap detector, and drug safety. PIN: 1234
Mobile BYOD Portal	http://localhost:5173/mobile.html	Smartphone-optimized interface for QR code token generation.
FastAPI Interactive Docs	http://localhost:8000/docs	Interactive Swagger UI for testing all REST endpoints and schemas.
Backend Health Check	http://localhost:8000/api/health	JSON status endpoint confirming active LLM provider and DB health.

7. Troubleshooting & Common Fixes

- **Port Conflict (8000 / 5173 in use):** Kill existing processes.
 - macOS: `lsof -ti:8000,5173 | xargs kill -9`
 - Windows: `Get-Process -Id (Get-NetTCPConnection -LocalPort 8000).OwningProcess | Stop-Process`
- **Ollama Model Not Found:** Run `ollama list` and make sure `qwen2.5:7b` is downloaded.
- **Microphone / Audio Permission:** Browser requires HTTPS or localhost to access microphone API.
- **Doctor Portal Sign-in:** The default security PIN for doctor encounter sign-off is **1234**.